Design, Simulation and Physical Verification of a High-Swing Cascode Current Mirror

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***Abstract*—This paper presents the design and simulation of a high-swing (wide-swing) cascode current mirror that overcomes the low output resistance of a simple current mirror and the high voltage-headroom requirement of a conventional cascode mirror, while maintaining stable 100 µA operation at a 1.8 V supply. The circuit is implemented in Cadence Virtuoso with the GPDK090 library using five NMOS transistors (mirror devices W = 12 µm, L = 1 µm; bias device W = 3 µm = W/4), with the cascode gates biased at $V_{TH} + 2V_{ov}$ by the quarter-width diode device. Simulation shows an output current of 100.06 µA, an output resistance of 1.57 MΩ (13× the simple mirror's 0.12 MΩ), and 99% current compliance at only 0.44 V of output voltage — 0.35 V less headroom than the conventional cascode's 0.79 V. A full custom layout was drawn and verified: Diva DRC reports zero errors and LVS matches the schematic exactly. A side-by-side sweep of all three mirror topologies quantifies the trade-offs.**

***Index Terms*—Cascode current mirror, wide-swing, high-swing, output resistance, voltage headroom, Cadence Virtuoso, GPDK090, DRC, LVS.**

I. INTRODUCTION

A simple two-transistor current mirror has an output resistance of only $r_o = 1/\lambda I_D$, so its output current varies appreciably with output voltage — a serious limitation when the mirror biases high-gain stages [1], [2]. Cascoding multiplies the output resistance by $g_m r_o$, but the conventional cascode connection, which biases the cascode gate from a stacked diode pair, wastes headroom: its minimum output voltage is $V_{TH} + 2V_{ov}$, a full threshold voltage more than the physical limit of two saturation voltages.

The high-swing (wide-swing) cascode mirror, attributed to Sooch [3], removes this waste. A separate bias device of width W/4 — carrying the same reference current at twice the overdrive — generates $V_B \approx V_{TH} + 2V_{ov}$ for the cascode gates, while the mirror gates are driven from the input node. The bottom devices then sit at $V_{DS} \approx V_{ov}$, the edge of saturation, and the output complies down to about $2V_{ov}$. This work designs the circuit at 100 µA and 1.8 V in GPDK090, verifies it against the BSIM3v3 models, compares all three topologies in one simulation, and carries the design through layout, DRC and LVS.

II. CIRCUIT CONFIGURATION AND DESIGN

A. Topology

The circuit (Fig. 1) contains five NMOS devices. M0 (W = 3 µm) is diode-connected and fed by I_0 = 100 µA; its gate voltage V_B drives the gates of the cascode devices M3 and M4. The input branch stacks cascode M3 on mirror device M1; the gates of M1 and M2 are tied to the input node (the wide-swing connection), not to M1's drain. The output branch stacks M4 on M2. All mirror and cascode devices use W = 12 µm, L = 1 µm. The output is exercised by a swept/AC source V2.

TABLE I
DESIGN SPECIFICATIONS

Parameter	Value
Supply voltage, VDD	1.8 V
Reference currents I0, I1	100 µA each
Mirror/cascode devices M1–M4	W = 12 µm, L = 1 µm
Bias device M0	W = 3 µm = W/4, L = 1 µm
Technology	GPDK090, nmos2v, BSIM3v3
Output test source	DC sweep + 100 mV, 1 kHz sine

B. Bias design

With $I_D = \tfrac{1}{2}\mu_n C_{ox}(W/L)V_{ov}^2$, quartering W at constant current doubles the overdrive. M0 therefore develops $V_{GS0} = V_{TH} + 2V_{ov}$, which is exactly the cascode-gate bias that places the bottom devices at $V_{DS} = V_B - V_{GS3} \approx V_{ov}$. Simulation gives V_B = 1.121 V, and the bottom devices indeed settle at $V_{DS} \approx$ 0.251 V against V_{Dsat} = 0.24 V — the edge of saturation, as intended. The ideal minimum output voltage is $2V_{ov} \approx$ 0.48 V, versus $V_{TH} + 2V_{ov} \approx$ 0.96 V for the conventional cascode.

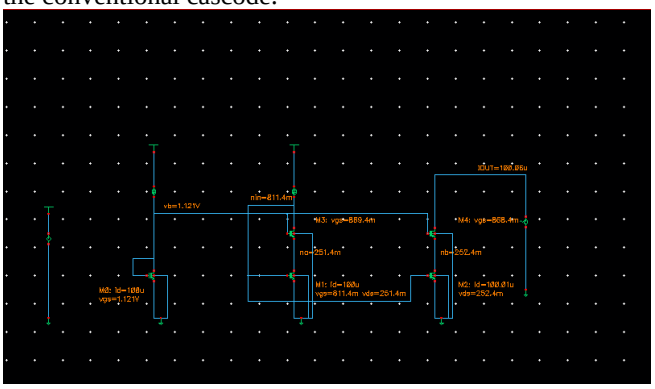


Fig. 1. Cadence Virtuoso schematic with the simulated DC operating point annotated (V_B = 1.121 V, node na = 251 mV, I_{OUT} = 100.06 µA).

III. SIMULATION METHODOLOGY

The schematic was simulated with Spectre (NN corner). The DC operating point was verified device by device; the output voltage was swept 0–1.8 V to measure compliance;

an AC analysis on the output source measured the small-signal output impedance; a transient analysis applied a 100 mV, 1 kHz sine at the output; and the supply was swept 0–1.8 V. In a second deck, simple, conventional-cascode and high-swing mirrors — all from identical devices at 100 µA — share one swept output variable so the three compliance curves superimpose exactly (Fig. 2).

IV. RESULTS AND DISCUSSION

A. Operating point

All five devices operate in saturation: M0 at V_{GS} = 1.121 V (V_{ov} = 0.645 V ≈ 2× the mirror overdrive), M1/M2 at V_{GS} = 811 mV with V_{DS} ≈ 251 mV, and the cascodes at V_{GS} ≈ 869 mV. The output current is 100.06 µA at V_{OUT} = 0.9 V — a 0.06% mirroring error.

B. Topology comparison

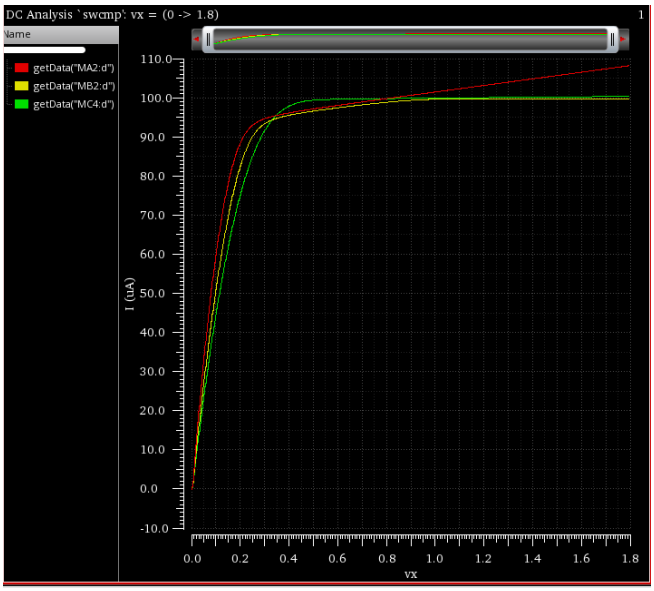


Fig. 2. Output current vs output voltage for the simple mirror (red), conventional cascode (yellow) and high-swing cascode (green), all at 100 µA. The high-swing curve is both flat and early.

TABLE II
MEASURED COMPARISON OF THE THREE TOPOLOGIES

Metric	Simple	Conv. cascode	High-swing
VOUT for 95% IOUT	0.31 V	0.35 V	0.34 V
VOUT for 99% IOUT	0.68 V	0.79 V	0.44 V
Rout (slope 1–1.7 V)	0.12 MΩ	4.67 MΩ	1.79 MΩ
IOUT error at 1.8 V	+8.4%	+0.001%	+0.5%

The simple mirror reaches 95% quickly but never flattens — its current is 8.4% high at 1.8 V and its output resistance is only 0.12 MΩ. The conventional cascode is extremely flat but needs 0.79 V before reaching 99% of the target. The high-swing mirror achieves 99% at just 0.44 V — saving 0.35 V ≈ V_{TH} of headroom — while raising output resistance 13× over the simple mirror. Its R_{out} is lower than the conventional cascode's because its bottom devices operate at the saturation edge (V_{DS} ≈ V_{ov}), where g_{ds} is larger — the classic swing-versus-gain trade, bought deliberately for the headroom.

C. Output compliance and output resistance

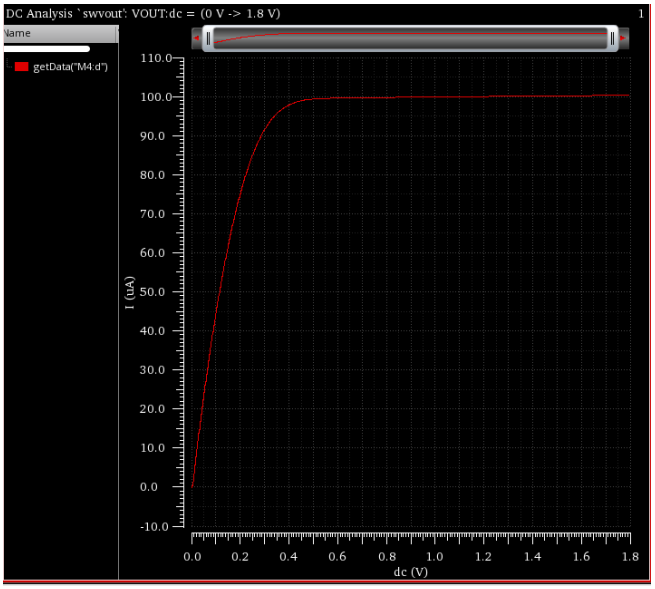


Fig. 3. I_D(M4) versus swept output voltage: 92% at 0.3 V, 98% at 0.4 V, 99.8% at 0.6 V.

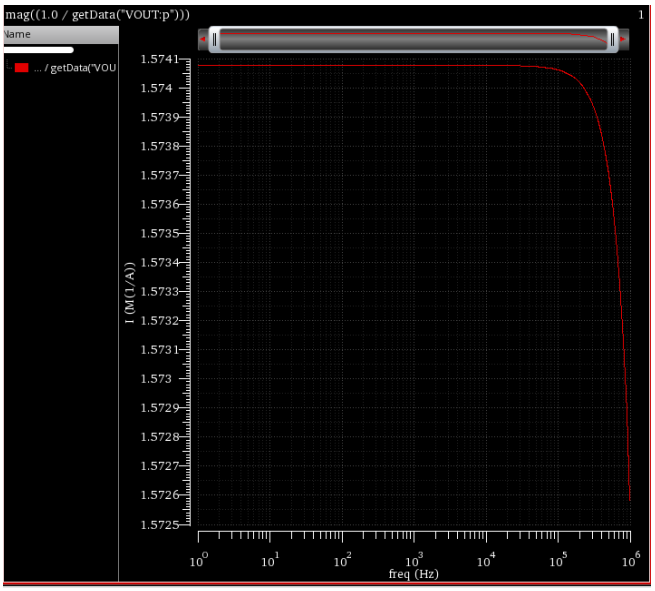


Fig. 4. Small-signal output impedance |V/I| from AC analysis: 1.574 MΩ, flat to ≈100 kHz.

The AC measurement (Fig. 4) gives R_{out} = 1.574 MΩ at 1 kHz, in agreement with the small-signal estimate $r_{o4}(1 + g_{m4}r_{o2})$ ≈ 1.3 MΩ and with the DC slope (1.79 MΩ). The roll-off above 100 kHz is the output capacitance of M4.

D. Transient behaviour

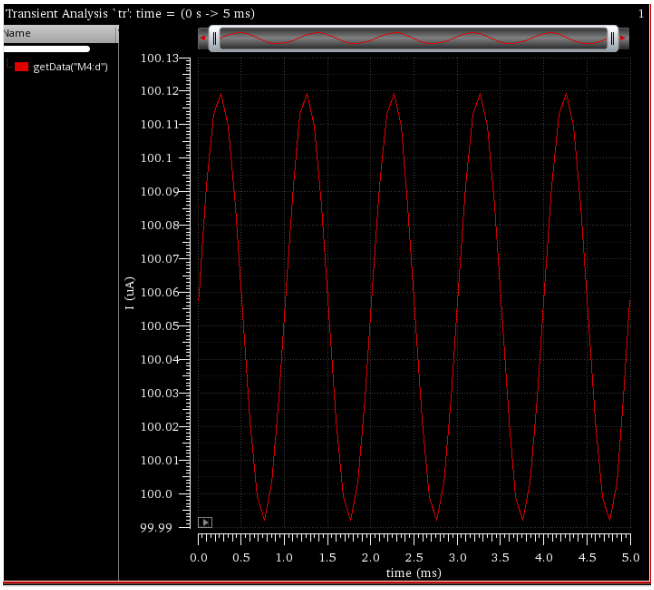


Fig. 5. Transient output current with a 100 mV, 1 kHz sine on the output node: 127 nA peak-to-peak ripple about 100.06 µA.

With a 100 mV-peak disturbance on the output node the mirrored current ripples only 127 nA peak-to-peak (0.13%), equivalent to R_{out} = 0.2 V/127 nA = 1.57 MΩ — confirming the AC result in the time domain and demonstrating stable current operation.

E. Supply sweep

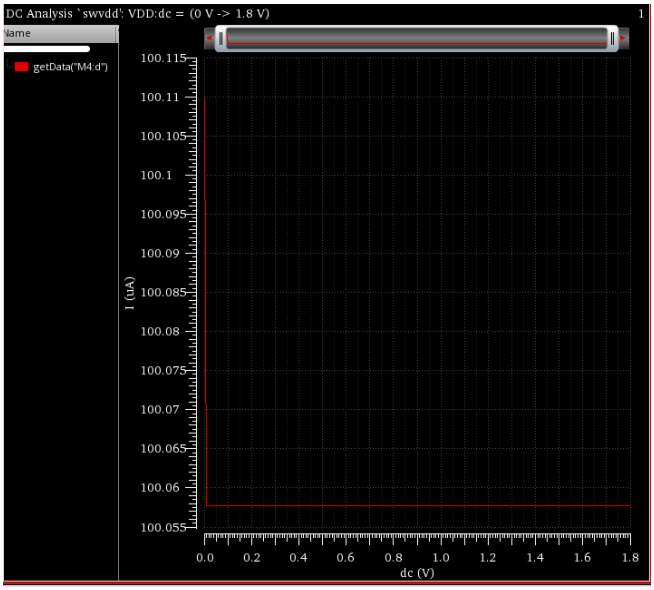


Fig. 6. Output current versus supply voltage at V_{OUT} = 0.9 V: with the reference branches biased, I_{OUT} varies less than 0.06% over the full sweep — supply-insensitive operation.

V. LAYOUT AND PHYSICAL VERIFICATION

The five-transistor core was laid out in ≈17 × 31 µm (Fig. 7): M0, M1, M2 in the bottom row with integrated body ties (all sources grounded), M3 and M4 stacked above with no local taps (their sources sit at ≈251 mV; bulk reaches ground through the substrate). M3 uses the pcell's switchSD option so its source column lands directly over M1's drain column, making the na connection a single vertical strap (Fig. 8). The gate bus of M1/M2 (the IIN net) crosses the na strap on a short Metal2 bridge with Via1 contacts — the only use of a second metal; everything else is Metal1. Diva DRC reports zero errors; extraction recognizes all five devices; and LVS declares "The net-lists match" with zero unmatched instances, nets, terminals, and zero size errors (Table III).

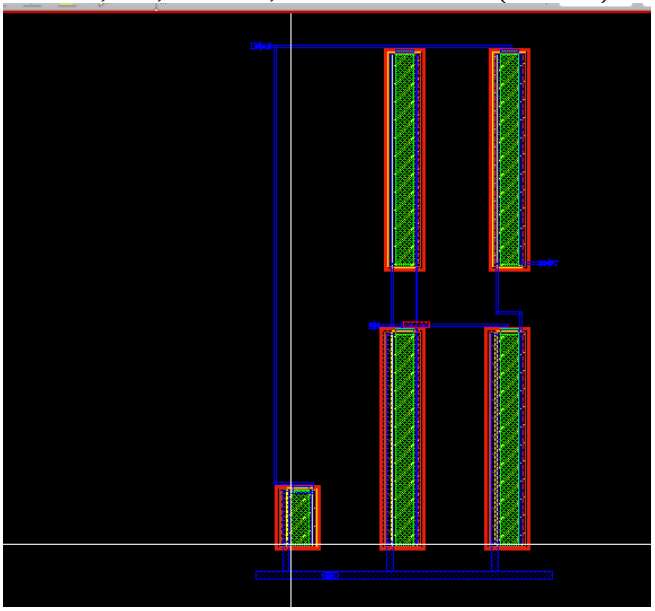


Fig. 7. Complete layout: bottom row M0 (W = 3 µm), M1, M2; top row cascodes M3, M4; V_B bus at top, GND rail at bottom.

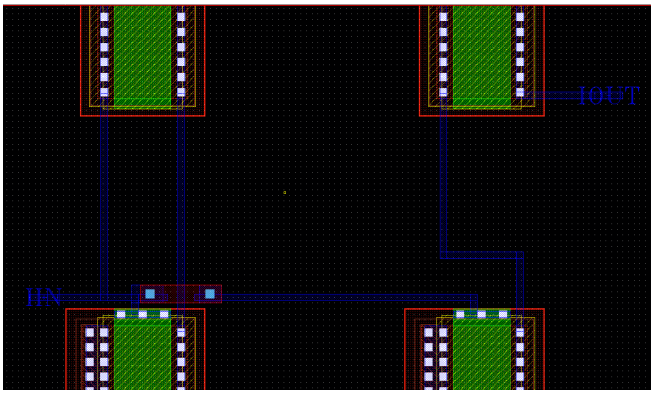


Fig. 8. Mid-region detail: the vertical na/nb straps between rows, the IIN gate bus, and the Metal2 bridge with Via1 pads.

TABLE III
PHYSICAL VERIFICATION SUMMARY

Check	Result
Diva DRC	0 errors
Extraction	5 nmos2v devices, 0 errors
LVS verdict	"The net-lists match", 0 size errors

VI. ADVANTAGES, LIMITATIONS, AND APPLICATIONS

The high-swing cascode delivers cascode-class output resistance at nearly the theoretical minimum compliance voltage, making it the standard mirror for low-voltage

analog design — op-amp loads, bias distribution and DAC current sources. Its costs are one extra bias branch (here 100 µA; in practice the bias device can be scaled to a fraction of the reference), sensitivity of the $V_{TH}+2V_{ov}$ bias to process and body effect (the W/4 rule is exact only for ideal square-law devices — margin must be verified against the technology models, as done here), and a somewhat lower R_{out} than the conventional cascode because the bottom devices sit at the saturation edge.

VII. CONCLUSION

A high-swing cascode current mirror was designed in Cadence Virtuoso with the GPDK090 library and verified against BSIM3v3 models at 100 µA and 1.8 V. The quarter-width bias device sets the cascode gates at $V_{TH} + 2V_{ov} = 1.121$ V, placing the mirror devices at the saturation edge. Measured against the simple and conventional-cascode mirrors in one simulation, the design achieves $R_{out} = 1.57$ MΩ (13× the simple mirror) with 99% compliance at 0.44 V (0.35 V better than the conventional cascode), and holds its current within 0.13% under a 100 mV output disturbance. The layout passes DRC cleanly and matches the schematic in LVS, completing the design through physical verification.

REFERENCES

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[3] N. S. Sooch, “MOS Cascode Current Mirror,” U.S. Patent 4,550,284, Oct. 29, 1985.